Gurugram University Scheme of Studies and Examination
Bachelor of Technology (Electronics and Communication Engineering)

SEMESTER VI

S. No.	Category	Course Code	Course Title	Hours per week			Credits	Marks for Sessional	Marks for End Term Examination	Total
				L	T	P				
1	PCC		Control System Engineering	3	0	0	3	30	70	100
2	PCC		Microwave & Radar Engineering	3	0	0	3	30	70	100
3	PEC		Program Elective - II	3	0	0	3	30	70	100
4	PEC		Program Elective - III	3	0	0	3	30	70	100
5	OEC		Open Elective - II	3	0	0	3	30	70	100
6	PCC		VLSI System Design	3	0	0	3	30	70	100
7	LC		Control System LAB	0	0	2	1	50	50	100
8	LC		Microwave & Radar Engineering Lab(P)	0	0	2	1	50	50	100
9	PROJ		Project-I	-	-	4	2	50	50	100
10	MC		Economics for Engineers	2	0	0	0	30	70	100*
Total				28			22	330	570	900

NOTE:

1. Economics for Engineers: The examination of the regular students will be conducted by the concerned college/Institute internally. Each student will be required to score a minimum of 40% marks to qualify in the paper. The marks will not be included in determining the percentage of marks obtained for the award of a degree.
2. At the end of the 6th semester, each student has to undergo Practical Training of 4/6 weeks in an Industry/ Institute/ Professional Organization/ Research Laboratory/ training center etc. and submit the typed report along with a certificate from the organization and its evaluation shall be carried out in the 7th Semester.
3. Choose any one from each of the Professional Elective Course-II and III
4. Choose any one from Open Elective Course-II

PROFESSIONAL ELECTIVE- II (Semester-VI)

Sr. No	Code	Subject	Credit
1.		Robotics & Automation	3
2.		Wireless and Sensor Networks	3
3.		Mobile Communications	3
4.		Power Electronics	3

PROFESSIONAL ELECTIVE - III (Semester-VI)

Sr. No	Code	Subject	Credit
1.		Nano electronics	3
2.		High Speed Electronics	3
3.		Biosensors	3
4.		Image Processing	3

Control System Engineering

Course Code					
Category	**Professional Core Courses**				
Course title	**Control System Engineering**				
Scheme	**L**	**T**	**P**	**Credits**	**Semester : VI**
	3	**1**	**0**	**3**	
Class Work	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	3Hrs				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:
1. To understand concepts of the mathematical modeling, feedback control and stability analysis in Time and Frequency domains
2. To develop skills, to analyze feedback control systems in continuous- and discrete time domains.
3. To learn methods for improving system response transient and steady state behavior (response).
4. The compensator design of linear systems is also introduced.

UNIT-I

Systems Components and Their RepresentationControl System: Terminology and Basic Structure-Feed forward and Feedback control theory-Electrical and Mechanical Transfer Function Models-Block diagram Models-Signal flow graphsmodels-DC and AC servo Systems-Synchronous -Multivariable control system

UNIT-II

Time Response Analysis and Stability ConceptTransient response-steady state response-Measures of performance of the standard first orderand second order system-effect on an additional zero and an additional pole-steady errorconstant and system- type number-PID control.
Concept of stability-Bounded - Input Bounded - Output stability-Routh stability criterion-Relative stability-Root locus concept-Guidelines for sketching root locus.

UNIT-III

Frequency Domain AnalysisBode Plot - Polar Plot- Nyquist Plots-Design of compensators using Bode Plots-Cascade leadcompensation-Cascade lag compensation-Cascade lag-lead compensation

UNIT-IV

Control System Analysis Using State Variable MethodsState variable representation-Conversion of state variable models to transfer functions-Conversion of transfer functions to state variable models-Solution of state equations-Concepts ofControllability and Observability-Stability of linear systems-Equivalence between transferfunction and state variable representations

Course Outcomes: At the end of this course students will demonstrate the ability to
1. Understand the concepts of control systems and importance of feedback in control systems.
2. Perform signal flow graph and formulate transfer function.
3. Perform computations and solve problems on frequency response analysis.
4. Analyse Polar, Bode and Nyquist's plot.
5. Evaluate different types of state models and time functions.
6. Analyse different types of control systems like linear and non-linear control systems, etc.

Text/Reference Books:

1. **B.S Manke , Linear Control system, Khanna Publication**
2. Gopal. M., "Control Systems: Principles and Design", Tata McGraw-Hill, 1997
3. Ambikapathy A., Control Systems, Khanna Book Publications, 2019.
4. Kuo, B.C., "Automatic Control System", Prentice Hall, sixth edition, 1993.
5. Ogata, K., "Modern Control Engineering", Prentice Hall, second edition, 1991.
6. Nagrath & Gopal, "Modern Control Engineering", New Age International, New Delhi

MICROWAVE AND RADAR ENGINEERING

Course Code					
Category	**Professional Core Courses**				
Course title	**Microwave and Radar Engineering**				
Scheme	**L**	**T**	**P**	**Credits**	**Semester : VI**
	3	**1**	**0**	**3**	
Class Work	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	3Hrs				

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To build up the concept from basics of microwave communications to modern applications.
2. To analyze and study rectangular and circular wave guides using field theory.
3. To understand the theoretical principles underlying microwave devices and networks.
4. To design microwave components such as power dividers, hybrid junctions, Directional Couplers, microwave filters, Microwave Wave-guides and Components, Ferrite Devices.
5. To study about Microwave Solid-State Microwave Devices and Microwave Tubes. 5.
6. To Study about Microwave Measurement Techniques.

UNIT I

Transmission Line: Transmission line equations & solutions, reflection and transmission coefficient, standing wave, standing wave ratio, line impedance and admittance, Introduction to strip lines, Microstrip Transmission line (TL). Wave Guide: Rectangular Wave guide -Field Components and Parameters, TE, TM Modes, Dominant Mode, Circular Waveguides: TE, TM modes. Wave Velocities, Wave guide Cavities.

UNIT II

Passive microwave devices: Microwave Junctions and Couplers, Scattering Matrix, Passive microwave devices: Microwave Hybrid Circuits, Terminations, Attenuators, Phase Shifters, Microwave Propagation in ferrites, Faraday Rotation, Isolators, Circulators. S parameter analysis of all components.

UNIT III

Microwave tubes : Microwave Tubes: Limitation of Conventional Active Devices at Microwave frequency, Two Cavity Klystron, Reflex Klystron, Magnetron, Traveling Wave Tube, Backward Wave Oscillators: Their Schematic, Principle of Operation, Performance Characteristic and their applications. Microwave Measurements: Measurement of Insertion Loss, Frequency, Cavity Q, Dielectric Constant, Scattering Parameters, Noise Factors, Return Loss, Impendence; VSWR Metering and Measurement, High Power Measurement; Power Meters, Microwave Amplifiers.

UNIT IV

Introduction to RADAR systems: RADAR Block diagram, RADAR Range equation, Probability of detection of false alarm, Integration of RADAR pulses, RADAR cross UNIT I of targets, MTI RADAR, CW RADAR.

Course Outcomes:

1. Analyze various parameters and characteristics of the transmission line and waveguide and also use of wave guide component as per applications.
2. Describe, analyze and design simple microwave circuits and devices e g couplers, Attenuators, Phase Shifter and Isolators.
3. Student will also understand the microwave propagation in ferrites.
4. Analyze the difference between the conventional tubes and the microwave tubes for the transmission of the EM waves.
5. Acquire knowledge about the handling and measurement of microwave equipment.
6. Differentiate different Radars, find applications and use of its supporting systems.

Text Books:

1. Liao, S.Y., " Microwave Devices & Circuits", 3rd Edition, Prentice Hall of India Publication, 1995.
2. Sushrut Das, "Microwave Engineering", 1st Edition, Oxford University Publication, 2015.
3. M.I. Skolnik, "Introduction to Radar Engineering ", 3rd Edition, Tata McGraw Hill Publication, 2001.**Reference Books:**

4.. A Das and S.K. Das, "Microwave Engineering", 1st Edition, Tata McGraw Hill Publication, 2000.

VLSI SYSTEM DESIGN

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Core Courses</td></tr>
<tr><td>Course title</td><td colspan="5">VLSI System Design</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester : VI</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">03Hrs</td></tr>
</table>

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To learn basic CMOS Circuits
2. To nurture students with CMOS analog circuit designs.
3. To realize importance of testability in logic circuit design.
4. To learn CMOS process technology.
5. To learn the concepts of designing VLSI Subsystems.

UNIT I

Introduction to MOSFETs : MOS Transistor Theory – Introduction MOS Device, Fabrication and Modeling , Body Effect, Noise Margin; Latch-up MOS Inverter :

UNIT II

MOS Transistors, MOS Transistor Switches, CMOS Logic, Circuit and System Representations, Design Equations, Static Load MOS Inverters, Transistor Sizing, Static and Switching Characteristics; MOS Capacitor; Resistivity of Various Layers. Symbolic and Physical Layout Systems – MOS Layers Stick/Layout Diagrams; Layout Design Rules, Issues of Scaling, Scaling factor for device parameters.

UNIT III

Combinational MOS Logic Circuits: Pass Transistors/Transmission Gates; Designing with transmission gates, Primitive Logic Gates; Complex Logic Circuits. Sequential MOS Logic Circuits: SR Latch, clocked Latch and flip flop circuits, CMOS D latch and edge triggered flip flop. Dynamic Logic Circuits; Basic principle, non ideal effects, domino CMOS Logic, high performance dynamic CMOSCircuits

UNIT IV

Clocking Issues, Two phase clocking. CMOS Subsystem Design: Semiconductor memories, memory chip organization, RAM Cells, dynamic memory cell.

Course outcomes: At the end of this course students will demonstrate the ability to

1. Understand MOS transistor theory and short channel effects.
2. Calculate Noise Margins & Propagation Delay of CMOS Inverter.
3. Analyze the combinational CMOS circuit for speed, power & area.
4. Implement combinational & sequential CMOS circuit with various topologies like domino logic.
5. Design of memories with efficient architectures to improve access times, power consumption.
6. Design an application using CMOS.

TEXT / REFERENCE BOOKS:

1. S. M. Kang and Y. Leblebici,CMOS Digital Integrated Circuits : Analysis andDesign, Third Edition, MH, 2002.
2. W. Wolf,Modern VLSI Design : System on Chip, Third Edition, PH/Pearson, 2002.
3. N. Weste, K. Eshraghian and M. J. S. Smith,Principles of CMOS VLSI Design : A Systems Perspective, Second Edition (Expanded), AW/Pearson, 2001.
4. J. M. Rabaey, A. P. Chandrakasan and B. Nikolic,Digital Integrated Circuits : A Design Perspective, Second Edition, PH/Pearson, 2003.
5. D. A. Pucknell and K. Eshraghian,Basic VLSI Design : Systems and Circuits, Third Edition, PHI, 1994.

CONTROL SYSTEM LABORATORY

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Laboratory Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Control System Laboratory</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">02Hrs</td></tr>
</table>

Notes:

(iii) At least 10 experiments are to be performed by students in the semester.

(iv) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus. Group of students for practical should be 15 to 20 in number.

LIST OF EXPERIMENTS: ANY SIX EXPERIEMENTS

1. To study speed Torque characteristics of
   a) A.C. servo motor
   b) DC servo motor.
2. (a) To demonstrate simple motor driven closed loop DC position control system.
   (b) To study and demonstrate simple closed loop speed control system.
3. To study the lead, lag, lead-lag compensators and to draw their magnitude and phase plots.
4. To study a stepper motor & to execute microprocessor or computer-based control of the same by changing number of steps, direction of rotation & speed.
5. To implement a PID controller for temperature control of a pilot plant.
6. To study behavior of 1st order, 2nd order type 0, type 1 system.
7. To study control action of light control device.
8. To study water level control using a industrial PLC.
9. To study motion control of a conveyor belt using an industrial PLC

SOFTWARE BASED (ANY FOUR EXPT.)

Introduction to SOFTWARE (Control System Toolbox)

10. Different Toolboxes in SOFTWARE, Introduction to Control Systems Toolbox.
11. Determine transpose, inverse values of given matrix.
12. Plot the pole-zero configuration in s-plane for the given transfer function. Plot unitstep response of given transfer function and find peak overshoot, peak time.
13. Plot unit step response and to find rise time and delay time.
14. Plot locus of given transfer function, locate closed loop poles for different values ofk.
15. Plot root locus of given transfer function and to find out S, Wd, Wn at given root & todiscuss stability.
16. Plot bode plot of given transfer function and find gain and phase margins Plot the Nyquist plot for given transfer function and to discuss closed loop stability, gain and phase margin.

Note:

1. Each laboratory group shall not be more than about 20 students.
2. To allow fair opportunity of practical hands-on experience to each student, each experiment may either done by each student individually or in group of not more than 3-4 students. Larger groups be strictly discouraged/ disallowed.

Lab Outcomes: At the end of this lab students will demonstrate the ability to

1. Understand the concepts of control systems and importance of feedback in control systems.
2. Perform signal flow graph and formulate transfer function.
3. Perform computations and solve problems on frequency response analysis.
4. Analyse Polar, Bode and Nyquist's plot.
5. Evaluate different types of state models and time functions.
6. Analyse different types of control systems like linear and non-linear control systems, etc

MICROWAVE & RADAR ENGINEERING LABORATORY

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Laboratory Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Microwave & Radar Engineering Laboratory</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">02Hrs</td></tr>
</table>

Notes:

(i) At least 10 experiments are to be performed by students in the semester.

(ii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus. Group of students for practical should be 15 to 20 in number.

COURSE OBJECTIVE:

1. Know about the behavior of microwave components.
2. Understand the radiation pattern of horn antenna.

LIST OF EXPERIMENTS:

1. To study wave guide components.
2. To study the characteristics of Gunn oscillator Gun diode as modulated source.
3. Study of wave guide horn and its radiation pattern and determination of the beam width.
4. To study isolation and coupling coefficient of a magic Tee.
5. To measure coupling coefficient, Insertion loss & Directivity of a Directional coupler.
6. To measure attenuation and insertion loss of a fixed and variable attenuator.
7. To measure isolation and insertion loss of a three port Circulators/Isolator.
8. To measure the standing wave ratio and reflection coefficient in a Microwave Transmission line.
9. To measure the frequency of a microwave source and demonstrate relationship among guide dimensions, free space wavelength and guide wavelength.
10. To measure the impedance of unknown load.
11. Use Doppler RADAR to detect the maximum range.
12. Determine the velocity of the moving objects with the help of RADAR range.
13. Use RADAR system to measure the distance traveled by any object.

Course Outcomes:

1. Demonstrate the characteristics of Microwave sources.
2. Demonstrate the characteristics of directional Couplers
3. To test the characteristics of microwave components
4. To analyze the radiation pattern of antenna
5. To measure antenna gain
6. Practice microwave measurement procedures

ECONOMICS FOR ENGINEERS

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Mandatory Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Economics for Engineers</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>2</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examination of the regular students will be conducted by the concerned college/Institute internally. Each student will be required to score a minimum of 40% marks to qualify in the paper. The marks will not be included in determining the percentage of marks obtained for the award of a degree.

Course Objectives:

1. Acquaint the students to basic concepts of economics and their operational significance.
2. Acquaint students with market and its operation.
3. To stimulate the students to think systematically and objectively about contemporary economic problems.

UNIT-I

Definition of Economics- Various definitions, types of economics- Micro and Macro Economics, nature of economic problem, Production Possibility Curve, Economic laws and their nature, Relationship between Science, Engineering, Technology and Economic Development.

Demand- Meaning of Demand, Law of Demand, Elasticity of Demand- meaning, factors effecting it, its practical application and importance.

UNIT-II

Production- Meaning of Production and factors of production, Law of variable proportions, Returns to scale, Internal and external economies and diseconomies of scale.

Various concepts of cost of production- Fixed cost, Variable cost, Money cost,Real cost, accounting cost, Marginal cost, Opportunity cost. Shape of Average cost,Marginal cost, Total cost etc. in short run and long run.

UNIT-III

Market- Meaning of Market, Types of Market- Perfect Competition, Monopoly, Monopolistic Competition and Oligopoly (main features).

Supply- Supply and law of supply, Role of demand & supply in price determinationand effect of changes in demand and supply on prices.

UNIT-IV

Indian Economy- Nature and characteristics of Indian economy as underdeveloped, developing and mixed economy (brief and elementaryintroduction), Privatization - meaning, merits and demerits.Globalization of Indian economy - merits and demerits.Banking- Concept of a Bank, Commercial Bank- functions, CentralBank- functions, Difference between Commercial & Central Bank.

Course outcomes:

1. The students will able to understand the basic concept of economics.
2. The students will able to understand the basic concept of demand.
3. The student will able to understand the concept of production and cost.
4. The student will able to understand the concept of market.
5. The students will able to understand the basic concept of supply.
6. The student will able to understand the concept of privatization, globalization and banks.

References:

1. Jain T.R., Economics for Engineers, VK Publication.
2. Chopra P. N., Principle of Economics, Kalyani Publishers.
3. Dewett K. K., Modern economic theory, S. Chand.
4. H. L. Ahuja., Modern economic theory, S. Chand.
5. Dutt Rudar&Sundhram K. P. M., Indian Economy.
6. Mishra S. K., Modern Micro Economics, Pragati Publications.
7. Singh Jaswinder, Managerial Economics, dreamtech press.
8. A Text Book of Economic Theory Stonier and Hague (Longman's Landon).
9. Micro Economic Theory – M.L. Jhingan (S.Chand).
10. Micro Economic Theory - H.L. Ahuja (S.Chand).
11. Modern Micro Economics: S.K. Mishra (Pragati Publications).

PROJECT-I

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Project</td></tr>
<tr><td>Course title</td><td colspan="5">Project-I</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Course objectives:

1. To allow students to demonstrate skills learned during their course of study by asking them to deliver a product that has passed through the design, analysis, testing and evaluation
2. To encourage research through the integration learned in a number of courses.
3. To allow students to develop problem solvingskills.
4. To encourage teamwork.
5. To improve students' communication skills by asking them to produce both a professional report and to give an oral presentation and prepare a technical report.

The students are required to undertake institutional project work.

The final Viva voice of the institutional project work will be conducted by an external examiner and one external examiner appointed by the institute. External examiner will be from the panel of examiners submitted by the concerned institute approved by the board of studies in engineering and technology. Assessment of institutional project work will be based on seminar, viva-voice and report of institutional project work obtained by the student from the industry or institute.

The internal marks distribution for the students consists of 50 marks internally and 50 marks by an external examiner.

Course outcomes

On successful completion of the course students will be able to:

1. Demonstrate a sound technical knowledge of their selected project topic.
2. Undertake problem identification and formulation.
3. Design engineering formula to complex problems utilising a systems approach.
4. Researchand engineering project.
5. Communicate with engineers and the community at large in written and oral form.
6. Demonstrate the knowledge, skills and attitudes of a professional engineer.

PROFESSIONAL ELECTIVE-II

ROBOTICS & AUTOMATION

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Robotics & Automation</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all UNITs, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objectives:

1. To learn about relationship between mechanical structures of industrial robots.
2. To gain understanding of spatial transformation to obtain forward kinematic equation of robot manipulators.
3. To understand inverse kinematics of simple robot manipulators.
4. To learn about Jacobian matrix and use it to identify singularities.

UNIT-I

Introduction: Concept and scope of automation: Socio economic impacts of automation, Types of Automation, Low-Cost Automation

Fluid Power: Fluid power control elements, Standard graphical symbols, Fluid power generators, Hydraulic and pneumatic Cylinders - construction, design and mounting; Hydraulic and pneumatic Valves for pressure, flow and direction control.

UNIT-II

Basic hydraulic and pneumatic circuits: Direct and Indirect Control of Single/Double Acting Cylinders, designing of logic circuits for a given time displacement diagram & sequence of operations, Hydraulic & Pneumatic Circuits using Time Delay Valve & Quick Exhaust Valve, Memory Circuit & Speed Control of a Cylinder, Troubleshooting and “Causes & Effects of Malfunctions” Basics of Control Chain, Circuit Layouts, Designation of specific Elements in a Circuit.

Fluidics: Boolean algebra, Truth Tables, Logic Gates, Coanda effect.

UNIT-III

Electrical and Electronic Controls: Basics of Programmable logic controllers (PLC), Architecture & Components of PLC, Ladder Logic Diagrams

Transfer Devices and feeders: Classification, Constructional details and Applications of Transfer devices, Vibratory bowl feeders, Reciprocating tube, Centrifugal hopper feeders

UNIT-IV

Robotics: Introduction, Classification based on geometry, control and path movement, Robot Specifications, Robot Performance Parameters, Robot Programming, Machine Vision, Teach pendants, Industrial Applications of Robots

Course Outcomes (COs): After studying this course, students will be able:

1. To demonstrate knowledge of the relationship between mechanical structures of industrial robots and
2. To learn robot’s operational workspace characteristics.
3. To demonstrate an ability to apply spatial transformation to obtain forward kinematic equation of robot manipulators.
5. To learn PLC
6. To demonstrate an ability to solve inverse kinematics of simple robot manipulators.
7. To demonstrate an ability to obtain the Jacobian matrix and use it to identify singularities.

Text Books:

1. Anthony Esposito, Fluid Power with applications, Pearson
2. S. R Majumdar, Pneumatic Control, McGraw Hill
3. S. R Deb, Robotic Technology and Flexible Automation, Tata Mc Hill
4. Saeed B. Niku Introduction to Robotics, Wiley India
5. Ashitava Ghosal, Robotics, Oxford

WIRELESS SENSOR NETWORKS

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Wireless sensor networks</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objectives:
1. Understand the working principles of the Sensors.
2. Understand the protocols used in sensor networks.
3. Understand design principles of WSN.
4. Understand engineering sensor networks.

UNIT-I
Introduction to Sensor Networks, unique constraints and challenges, Advantage of SensorNetworks, Applications of Sensor Networks, Types of wireless sensor networks, Mobile AdhocNetworks (MANETs) and Wireless Sensor Networks, Enabling technologies for WirelessSensor Networks. Issues and challenges in wireless sensor networks.

UNIT-II
Routing protocols, MAC protocols: Classification of MAC Protocols, S-MAC Protocol, BMACprotocol, IEEE 802.15.4 standard and ZigBee

UNIT-III
Dissemination protocol for large sensor network. Data dissemination, data gathering, and data fusion; Quality of a sensor network; Real-time traffic support and security protocols.Design Principles for WSNs, Gateway Concepts Need for gateway, WSN to Internet Communication, and Internet to WSN Communication

UNIT-IV
Single-node architecture, Hardware components & design constraints, Operating systems andexecution environments, introduction to TinyOS and nesC

Course Outcomes (COs): After studying this course, students will be able:
1. Design wireless sensor networks for a given application
2. Understand emerging research areas in the field of sensor networks
3. Understand MAC protocols used for different communication standards used in WSN
4. Understand large sensor network.
5. Understand architecture and hardware components.
6. Explore new protocols for WSN

Text Books:
1. WaltenegusDargie , Christian Poellabauer, “ Fundamentals Of Wireless Sensor Networks
2. Theory And Practice”, By John Wiley & Sons Publications ,2011
3. SabrieSoloman, “Sensors Handbook" by McGraw Hill publication. 2009
4. Feng Zhao, Leonidas Guibas, “Wireless Sensor Networks”, Elsevier Publications,2004
5. Kazem Sohrby, Daniel Minoli, “Wireless Sensor Networks”: Technology, Protocols andApplications, Wiley-Inter science
5. Philip Levis, And David Gay "TinyOS Programming” by Cambridge UniversityPress 2009

MOBILE COMMUNICATION

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Mobile Communication</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objectives:

During the course, students will be made to learn to:

1. Understand the Cellular concepts.
2. Understand the digital modulation techniques.
3. Understand the mobility in Cellular Systems.
4. Understand GSM.

UNIT-I

Cellular concepts- Cell structure, frequency reuse, cell splitting, channel assignment,handoff, interference, capacity, power control; Wireless Standards: Overview of 2G and 3Gcellular standards.

UNIT-II

Large scale signal propagation. Fading channels-Multipath and small-scale fading- Dopplershift, doppler spread, average and rms delay spread, coherence bandwidth and coherencetime, flat and frequency selective fading, slow and fast fading, average fade duration and
level crossing rate.
Okumura Model, Hata Model, PCS Extension to Hata Model, Walfisch and Bertoni Model,Wideband PCS Microcell Model, Indoor Propagation Models-Partition losses (Same Floor),Partition losses between Floors, Log-distance path loss model.

UNIT-III

Multiple access schemes-FDMA, TDMA, CDMA and SDMA. Modulation schemes- BPSK,QPSK and variants, QAM, MSK and GMSK, multicarrier modulation, OFDM and OFDMA.

UNIT-IV

Mobility in Cellular Systems: The Gateway Concept, Measurement Reports, MobilityProcedures - Mobile IP: Basic Components, Tunneling
GSM: Architecture, – UMTS: Architecture, Basics of CDMA, - Introduction to LTE: History,Architecture - OFDM – Uplink and Downlink Communication in LTE.

Course Outcomes (COs): After studying this course, students will be able:

1. To understand the working principles of the mobile communication systems.
2. To understand the relation between the user features and underlying technology.
3. To analyze mobile communication systems for improved performance.
4. To understand multiple access schemes.
5. To analyze mobility in cellular systems.
6. To discuss GSM.

Text Books:

1. WCY Lee, Mobile Cellular Telecommunications Systems, McGraw Hill, 1990.
2. WCY Lee, Mobile Communications Design Fundamentals, Prentice Hall, 1993.
3. Raymond Steele, Mobile Radio Communications, IEEE Press, New York, 1992.
4. AJ Viterbi, CDMA: Principles of Spread Spectrum Communications, Addison Wesley,1995.
5. VK Garg & JE Wilkes, Wireless & Personal Communication Systems, Prentice Hall,1996.

POWER ELECTRONICS

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Power Electronics</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester : VI</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To introduce power semiconductor devices, their switching principles and applications.
2. To explain the operation of AC-DC uncontrolled and controlled rectifier, DC-DC converters and DC-AC inverters.
3. To analyse the power electronic switch based rectifier, converters and inverters.
4. To introduce hardware tools for implementation of power electronic circuits.

UNIT I

Characteristics of Semiconductor Power Devices: Thyristor, power MOSFET and IGBT- Treatment should consist of structure, Characteristics, operation, ratings, protections and thermal considerations. Brief introduction to power devices viz. TRIAC, MOS controlled thyristor (MCT), Power Integrated Circuit (PIC) (Smart Power), Triggering/Driver, commutation and snubber circuits for thyristor, power MOSFETs and IGBTs (discrete and IC based).Concept of fast recovery and schottky diodes as freewheeling and feedback diode.Standard Driver Circuit Schematics for MoSFETs and IGBTs.

UNIT II

Controlled Rectifiers: Single phase: Study of semi and full bridge converters for R, RL, RLE and level loads. Analysis of load voltage and input current- Derivations of load form factor and ripple factor, Effect of source impedance, Input current Fourier series analysis of input current to derive input supply power factor, displacement factor and harmonic factor.

Choppers: Quadrant operations of Type A, Type B, Type C, Type D and type E choppers, Control techniques for choppers – TRC and CLC, Detailed analysis of Type A chopper. Step up chopper. Multiphase Chopper

UNIT III

Single-phase inverters: Principle of operation of full bridge square wave, quasi-square wave, PWM inverters and comparison of their performance. Driver circuits for above inverters and mathematical analysis of output (Fourier series) voltage and harmonic control at output of inverter (Fourier analysis of output voltage). Filters at the output of inverters, Single phase current source inverter

UNIT IV

Switching Power Supplies: Analysis of fly back, forward converters for SMPS, Resonant converters - need, concept of soft switching, switching trajectory and SOAR, Load resonant converter - series loaded half bridge DC-DC converter.

Applications: Power line disturbances, EMI/EMC, power conditioners. Block diagram and configuration of UPS, salient features of UPS, selection of battery and charger ratings, sizing of UPS. Separately excited DC motor drive. P M Stepper motor Drive.

Course Outcomes: At the end of this course students will demonstrate the ability to

1. Learn how to analyze inverters and some basic applications.
2. Analyze and design SMPS, controlled rectifiers DC to DC converters. and, DC to AC inverters.
3. Learn and design DC to AC inverters, Charge controllers
4. Analyze typical industrial application requirements and build a solution with commercially available power electronic devices
5. Analyse the operation of DC-DC choppers.
6. Analyse the operation of voltage source inverters

Text /Reference Books:

1) P.S. Bimbhra, Power Electronics, Khanna Book Publishing, 2022.
2) M Singh, K Khanchandani, "Power Electronics" McGraw Hill Education, 2nd Ed., 2017
3) Muhammad H. Rashid, "Power electronics" Prentice Hall of India.
4) Ned Mohan, Robbins, "Power electronics", edition III, John Wiley and sons.
5) P.C. Sen., "Modern Power Electronics", edition II, S.Chand & Co.
6) V.R.Moorthi, "Power Electronics", Oxford University Press.
7) Cyril W., Lander," Power Electronics", edition III,McGraw Hill.
8) G K Dubey,S R Doradla,:Thyristorised Power Controllers",New Age International Publishers. SCR manual from GE, USA.

PROFESSIONAL ELECTIVE-III

NANO ELECTRONICS

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Nano Electronics</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To understand various aspects of nano-technology and the processes involved inmaking nano components and material.
2. Toleverage advantages of the nano-materials and appropriate use in solvingpractical problems.
3. To understand various aspects of nano-technology.
4. To understand the processes involved inmaking nano components and material.

Unit-I

Introduction to nanotechnology, applications of nano electronics. Basics of QuantumMechanics: Wave nature of particles and wave-particle duality, Pauli Exclusion Principle,wave functions and Schrodinger's equations, Density of States, Band Theory of Solids,Particle in a box Concepts

Unit-II

Shrink-down approaches: CMOS scaling: advantages and limitations. Nanoscale MOSFETs,FINFETs, Vertical MOSFETs, system integration limits (interconnect issues etc.)

Unit-III

Nanostructure materials, classifications of nanostructure materials, zero dimensional, onedimensional, two dimensional and three dimensional, properties and applicationsCharacterization techniques for nanostructured materials: SEM, TEM and AFM

Unit-IV

Nano electronics devices: Resonant Tunneling Diode, Coulomb dots, Quantum blockade, Singleelectron transistors, Carbon nanotube electronics, Band structure and transport, devices,applications, 2D semiconductors and electronic devices, Graphene, atomistic simulation

Course Outcomes:

At the end of this course, students will demonstrate the ability to:

1. Understand various aspects of nano-technology.
2. Understand processes involved inmaking nano components and material.
3. Leverage advantages of the nano-materials and appropriate use in solvingpractical problems.
4. Understand various aspects of nano-technology and
5. Understand the processes involved inmaking nano components and material.
6. Analyse Nano Electronic devices.

Text/ reference books:

1. G.W. Hanson, Fundamentals of Nanoelectronics, Pearson, latest edition
2. W. Ranier, Nanoelectronics and Information Technology (AdvancedElectronic Materialand Novel Devices), Wiley-VCH, latest edition
3. K.E. Drexler, Nanosystems, Wiley, latest edition
4. J.H. Davies, The Physics of Low-Dimensional Semiconductors, CambridgeUniversity Press, latest edition
5. C.P. Poole, F. J. Owens, Introduction to Nanotechnology, Wiley, latest edition

HIGH SPEED ELECTRONICS

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">High Speed Electronics</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:
1. To Study the high-speed electronics system.
2. To Understand Radio frequency amplifiers.
3. To analyse mixers.
4. Learn the fabrication process.

Unit-I

Transmission line theory (basics) crosstalk and nonideal effects; signal integrity: impact ofpackages, vias, traces, connectors; non-ideal return current paths, high frequency power delivery,methodologies for design of high-speed buses; radiated emissions and minimizing system noise.

Unit-II

Noise Analysis: Sources, Noise Figure, Gain compression, Harmonic distortion, Inter -modulation, Cross-modulation, Dynamic range.
Devices: Passive and active, Lumped passive devices (models), Active (models, low vs Highfrequency)

Unit-III

RF Amplifier Design, Stability, Low Noise Amplifiers, Broadband Amplifiers (and Distributed)Power Amplifiers, Class A, B, AB and C, D E Integrated circuit realizations, Cross-overdistortion Efficiency RF power output stages.
Mixers –Up conversion Down conversion, Conversion gain and spurious response. OscillatorsPrinciples.PLL Transceiver architectures.

Unit-IV

Printed Circuit Board Anatomy, CAD tools for PCB design, Standard fabrication, MicroviaBoards. Board Assembly: Surface Mount Technology, Through Hole Technology, ProcessControl and Design challenges.

Course Outcomes:
At the end of this course, students will demonstrate the ability to:
1. Study the high-speed electronics system.
2. Understand significance and the areas of application of high-speed electronics circuits.
3. Understand the properties of various components used in high-speed electronics.
4. Understand Radio frequency amplifiers.
5. Analyse Mixers.
6. Design High-speed electronic system using appropriate components.

Text/ reference books:

1. Stephen H. Hall, Garrett W. Hall, James A. McCall “High-Speed Digital System Design: AHandbook of Interconnect Theory and Design Practices”, August 2000, Wiley-IEEE Press.
2. Thomas H. Lee, “The Design of CMOS Radio-Frequency Integrated Circuits”,CambridgeUniversity Press, 2004, ISBN 0521835399.
1. Behzad Razavi, “RF Microelectronics”, Prentice-Hall 1998, ISBN 0-13-887571-5.
2. Guillermo Gonzalez, “Microwave Transistor Amplifiers”, 2nd Edition, Prentice Hall.
3. Kai Chang, “RF and Microwave Wireless systems”, Wiley.
4. R.G. Kaduskar and V.B.Baru, Electronic Product design, Wiley India, 2011

BIOSENSORS

Course Code					
Category	**Professional Elective Courses**				
Course title	**Biosensors**				
Scheme	**L**	**T**	**P**	**Credits**	**Semester: VI**
	3	**0**	**0**	**3**	
Class Work	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	3Hrs				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To understand the basic principles and classification of sensors and measurands.
2. To know the hardware and software of DAQ system and Electronic Interface systems
3. To understand how to measure various parameters and helps to design simple biomedical sensors.
4. To study about the sensor measurements for biological applications.

UNIT-I

Overview of biosensors and their electrochemistry: Molecular reorganization: Enzymes, Antibodiesand DNA, Modification of bio recognition molecules for Selectivity and sensitivity, Fundamentals ofsurfaces and interfaces

UNIT-II

Bioinstrumentation and bioelectronics devices: Principles of potentiometry and potentiometricbiosensors, Principles of amperometry and amperometric biosensors, Optical Biosensors based onFiber optics, Introduction to Chemometrics, Biosensor arrays; Electronic nose and electronic tongue.

UNIT-III

Iron-Selective Field-Effect Transistor (ISFET), Immunologically Sensitive Field Effect Transistor(IMFET). Fabrication and miniaturization techniques.

UNIT-IV

Sensor-to-Frequency Conversion Data-Acquisition Systems: Hardware and Software of DataAcquisition System (DAS), Electronic Interface, Integrated Sensors, Wireless integration. Smartsensor, Nano sensor.

Course Outcomes:

At the end of this course students will demonstrate the ability to

1. Understand the basic principles and classification of sensors and measurands.
2. Understand the hardware and software of DAQ system and Electronic Interface systems.
3. Understand how to measure various parameters and helps to design simple biomedical sensors.
4. Explain the concept of molecular reorganization, fundamentals of surfaces and interfaces.
5. Elucidate the principles of different types of biosensors
6. Understand sensor measurements for biological applications.

Text Books

1. Gardner, J.W., Microsensors, Principles and Applications, John Wiley and Sons(1994).
2. Kovacs, G.T.A., Micromachined Transducer Sourcebook, McGrawHill (2001).
3. Turner, A.P.F., Karube,I., and Wilson G.S., BiosensorsFundamentals and Applications, OxfordUniversity Press (2008)
4. Jon Cooper, Biosensors A Practical Approach, Bellwether Books
5. Manoj Kumar Ram, Venkat R, Bhethanabolta, Sensors for chemical and biological applications,CRC Press

IMAGE PROCESSING

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Image Processing</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:
1. To understand need for image transforms different types of image transforms and their properties.
2. Analyse image processing application and Machine vision.
3. Implementing image compression and spatial and frequency domain techniques of image compression.
4. To understand different feature extraction techniques.

UNIT-I

INTRODUCTION: Image Processing Fourier Transform and Z-Transform Causality and stability Toeplit and Circulate Metrics orthogonal and unitary Matrices and Kronecker product, Markov Processes KI Transform Mean square Estimates and Orthogonal Principles.

IMAGE SAMPLING QUANTIZATION: Band Limited Image Sampling Versus Replication, Reconstruction ofImage from samples Sampling Theorem, Sampling Theorem for Random Fields, Optimal Sampling, Nonrectangular Grid Sampling, Sampling Aperture, Display Aperture/ Interpolation Functions, Lagrange Interpolation Moire Effect. Image Quantization Uniform Optimal Quantizer, Properties of Mean Square Quantizer, Commander Design Visual Quantization

UNIT-II

IMAGE TRANSFORMS: Two Dimensional Orthogonal and Unitary Transforms and their properties. One-dimensional and Two Dimensional DFT Cosine and Sine Transforms.Hadamard, Slant, Harr and KL,Transforms and their properties, Approximation to KI Transforms.

IMAGE REPRESENTATION BY STOCHASTIC MODELS: One Dimensional Causal Models, AR and ARMA models, Non Causal Representation Spectral factorization, Image Decomposition.

UNIT-III

IMAGE ENHANCEMENT AND RESTORATION: Point Operation, Histogram Modeling, Spatial Operations, Transform Operations, Multispectral Image Enhancement. Image Observation Models, Inverse and Wiener filtering; FIR Wiener Filters, Filtering using Image Transform Causal Models and recursive filtering Maximum entropy restoration. Extrapolation of band limited signal.

UNIT-IV

IMAGE ANALYSIS AND IMAGE COMPRESSION: Spatial feature extraction, Edge detection and boundary extraction Boundary, region and moment representations structures, Texture, Image Segmentation, Reconstruction from Projections, Pixel Coding, Productive Techniques, Transform Coding Theory, Coding of Image, Coding of two-tone image.

Course Outcomes: At the end of this course students will demonstrate the ability to
1. Understand the need for image transforms different types of image transforms and their properties.
2. Develop any image processing application and understand the rapid advances in Machine vision.
3. Learn different techniques employed for the enhancement of images.
4. Learn different causes for image degradation and overview of image restoration techniques.
5. Understand the need for image compression and to learn the spatial and frequency domain techniques of image compression.
6. Learn different feature extraction techniques for image analysis and recognition.

Text Books:

1. Anil Jain, Digital Image Processing, PHI.
2. Gonzalez and Woods, Image Processing, Addison Wesley & Sons.
3. Digital Image Enhancement, Restoration and Compression, 4th Edition, SE Umbaugh, Taylor & Francis/CRC Press, 2023
4. Yao wang, Joem Ostarmann and Ya – quin Zhang, “Video processing and communication”,1st edition, PHI